

A Survey on RRAM Technology and MOS Integration

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Abstract

Resistive random-access memory (RRAM) is one of three serious candidates to replace embedded flash below 28 nm, alongside spin-transfer-torque magnetoresistive RAM (STT-MRAM) and phase-change memory (PCM). This paper covers what RRAM is, why the valence-change mechanism (VCM) became the dominant switching approach, and how the cell integrates into modern CMOS. The basic cell is a metal–insulator–metal (MIM) stack. VCM, which uses HfO_2 or TaO_x as the switching oxide, won out over PCM and conductive-bridging RAM (CBRAM) for reasons tied to process compatibility, thermal stability, and reset current—all explained in detail. In production, every RRAM cell is paired with a CMOS MOSFET in a 1-transistor–1-resistor (1T1R) configuration. The transistor handles current compliance during programming, cell addressing, and sneak-leakage suppression from unselected cells. We work through a sizing example from the literature on a 0.25 μm planar process and track how the constraints change on FD-SOI and FinFET. Reliability, retention, and disturb mechanisms are treated in a dedicated section. The paper closes with a look at analog in-memory computing as the most active forward-looking application.

Index Terms—Resistive random-access memory, RRAM, ReRAM, valence-change mechanism, oxygen vacancy, HfO_2 , TaO_x , embedded non-volatile memory, 1T1R, BEOL integration, FinFET, FD-SOI, in-memory computing.

I. INTRODUCTION

Embedded flash has been the standard non-volatile memory in microcontrollers, automotive ICs, and IoT chips for more than two decades. The basic idea is simple: a floating-gate or charge-trap MOSFET stores charge across a tunnel oxide of about 7–10 nm. The problem is that this tunnel oxide has a minimum thickness set by the ten-year retention requirement, and that minimum does not shrink with the logic node. On top of that, the high-voltage charge pumps needed to program flash grow in area at advanced nodes. By 28 nm, embedded flash had become a process burden that foundries were not willing to carry forward. TSMC stopped offering it beyond 28 nm in the late 2010s [5], and the gap it left created the market for emerging NVM.

Three technologies stepped in: STT-MRAM, RRAM, and PCM [5]. STT-MRAM is the most established at FinFET-class nodes, with shipping macros at TSMC 22ULL/16 nm, Samsung 28FDS, and GlobalFoundries 22FDX. PCM had its biggest commercial moment in Intel–Micron 3D XPoint (Optane), which ran from 2017 to 2022, and is currently in embedded production at STMicroelectronics on a 28 nm FD-SOI automotive platform [5]. RRAM is most established at planar bulk and FD-SOI nodes, with products at TSMC 40/28/22 nm [7] and at GlobalFoundries via Weebit Nano. Intel also demonstrated embedded RRAM in the 22FFL FinFET process using a TaO_x switching layer [8].

RRAM is attractive for practical reasons. The cell is a simple two-terminal MIM stack made from materials already used in CMOS fabrication. It goes in the back-end-of-line (BEOL), above the transistors, so it does not touch the front-end logic flow. The mask adder is small. In production, every cell is paired with a select MOSFET in a 1T1R configuration, which connects RRAM directly to MOSFET device modeling—the topic of this course.

This paper is organized around four questions mirroring our project presentation. Section II introduces the RRAM cell and main switching mechanisms, focusing on VCM. Section III explains why RRAM is being adopted and why VCM specifically won out. Section IV covers 1T1R integration and the select-FET sizing problem. Section V addresses reliability, retention, and disturb mechanisms. Section VI compares planar bulk, FD-SOI, and FinFET as host architectures. Section VII closes with an outlook on in-memory computing.

II. WHAT IS RRAM?

A. The RRAM Cell

RRAM stores information in the resistance of a two-terminal device [1, 2]. High resistance (HRS) is logic 0; low resistance (LRS) is logic 1. The transition from HRS to LRS is called SET. The reverse is RESET. Because the storage element is a resistor rather than a charge-storing capacitor, there is no tunnel-oxide thickness floor to scale around.

In an array, the resistive switching element (RS) needs a non-linear element (NLE) in series to select the right cell. Fig. 1 shows the two options: a two-terminal selector, or a CMOS MOSFET. In practice, the MOSFET option—the 1T1R cell—is what every embedded RRAM product uses today [2, 3]. One transistor handles addressing, current compliance during programming, and leakage isolation simultaneously.

B. Resistive Switching Mechanisms

Many physical mechanisms can produce a reversible resistance change in a thin film. Waser *et al.* organize the full landscape into the taxonomy shown in Fig. 2, covering nanomechanical, molecular, electrostatic/electronic, electrochemical metallization, valence-change, thermochemical, phase-change, magnetoresistive, and ferroelectric tunneling effects [2]. The right-hand axis classifies each by dominant species and by whether switching is

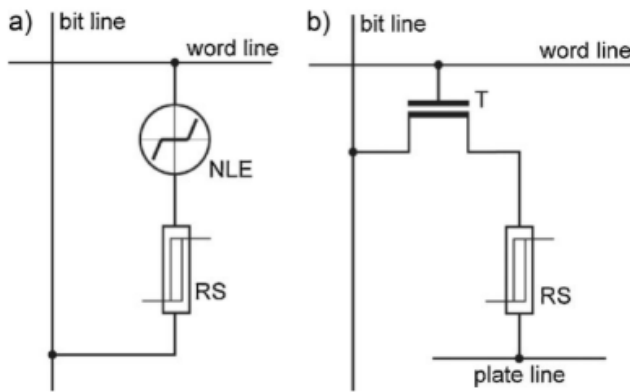


Fig. 1. RRAM cell architectures. (a) Selector-based cell, with a two-terminal NLE in series with the resistive switch (RS). (b) 1T1R cell, in which a CMOS MOSFET serves as the select element. Reproduced from [2].

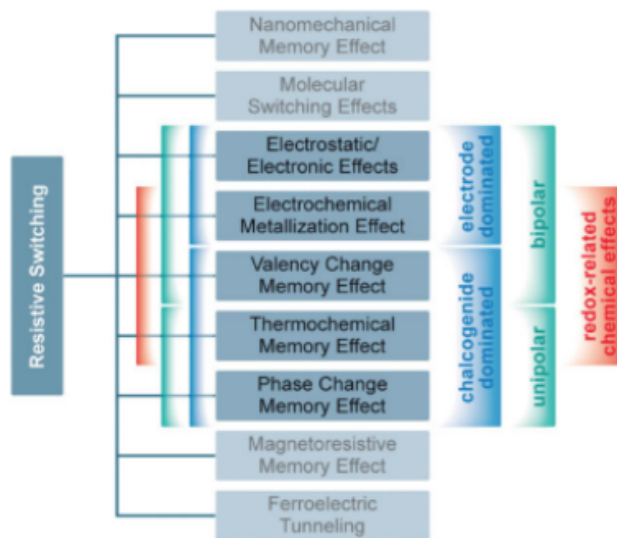


Fig. 2. Taxonomy of resistive switching mechanisms. Reproduced from [2].

unipolar or bipolar.

Three redox-based mechanisms matter most for memory.

Electrochemical Metallization Mechanism (ECM). An active electrode—usually Ag or Cu—dissolves under positive bias, drifts through a solid electrolyte, and forms a metallic filament bridging the two electrodes [2]. ECM can switch at low voltages (as low as ~ 0.2 V in Ag/GeS systems; typical commercial CBRAM windows are 0.5–2 V), but the metallic filament is thermally sensitive, which causes retention problems.

Thermochemical Mechanism (TCM). Joule heating drives switching. SET and RESET use the same polarity but different current levels—unipolar. NiO is the canonical TCM material [2]. TCM is commercially dead because each RESET is essentially a controlled local melt of the filament, and the resulting cycle-to-

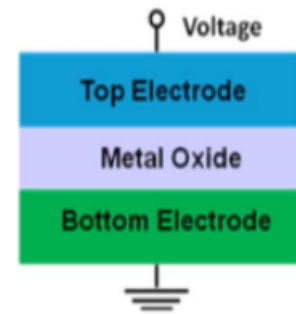


Fig. 3. The metal–insulator–metal (MIM) cell stack. A transition-metal oxide is sandwiched between two metal electrodes. Reproduced from [1].

cycle variability cannot be managed.

Valence-Change Mechanism (VCM). The mobile species is the oxygen anion, or equivalently the oxygen vacancy V_O . Under an applied field, oxygen drifts away from part of the oxide, leaving an oxygen-poor, sub-stoichiometric conducting path. The metal cations in that path are reduced to a lower oxidation state ($Hf^{4+} \rightarrow Hf^{3+}$, $Ta^{5+} \rightarrow Ta^{4+}$)—hence the name [2]. Reversing the bias drives oxygen back. VCM is bipolar and covers HfO_x and TaO_x RRAM [1, 3].

VCM with HfO_2 or TaO_x is what the industry settled on. The reasons are explained in the comparison with PCM and CBRAM in Section III-B, but briefly: HfO_2 is already the high- κ gate dielectric in HKMG MOSFETs, so deposition equipment and contamination protocols are already in the fab, and the oxygen-vacancy filament is more thermally stable than what ECM produces.

C. How VCM Works

A VCM cell is a passive MIM stack: a transition-metal oxide a few nanometers thick sits between a top and bottom electrode (Fig. 3). Applying a voltage drives oxygen ions and vacancies to migrate and organize into a conductive filament connecting the two electrodes. Reducing or reversing the voltage partially breaks that filament, returning the cell to HRS [1, 3].

The full cycle is shown in Fig. 4. In the as-fabricated state the oxide is essentially an insulator—the initial resistance state (IRS). Before the cell can operate, it goes through a one-time step called *forming* or *electroforming*: a voltage higher than the normal SET voltage is applied, causing a soft dielectric breakdown that nucleates the filament for the first time [1]. During forming the current is clamped to a compliance level I_{CC} by the series transistor. This is critical—without it, the energy deposited into the dielectric would cause hard, permanent breakdown.

After forming, the cell operates at lower voltages. SET (HRS $\rightarrow$ LRS) reconnects the filament. RESET (LRS $\rightarrow$ HRS) partially ruptures it. In bipolar operation, SET and RESET use opposite polarity. In unipolar, they share polarity but use different current levels [1, 2].

In the LRS, conduction is approximately ohmic, dominated by the sub-stoichiometric filament. In the HRS, conduction is more

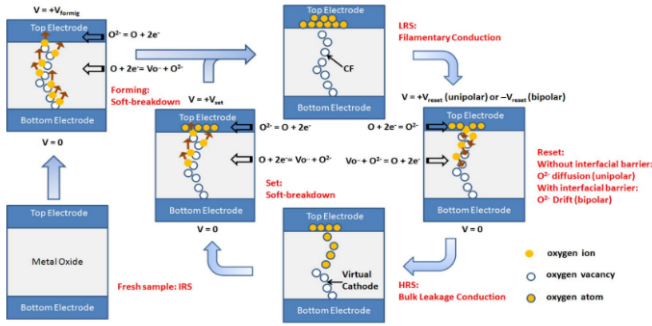


Fig. 4. Operation of a VCM cell. A forming pulse nucleates the oxygen-vacancy filament from the as-fabricated insulating state. Subsequent SET pulses reconnect the filament (LRS); RESET pulses partially rupture it (HRS). Reproduced from [1].

complex—trap-assisted tunneling, Poole–Frenkel emission, and Schottky emission at the oxide/electrode interface have all been reported depending on the material [1].

One important detail: RESET does *not* restore the oxide to its as-fabricated state. Forming happens once. Every SET and RESET after that reorganizes oxygen vacancies along the same path forming opened. The resistance depends on that accumulated history, not just on the instantaneous voltage—which is why RRAM belongs to the memristor family described by Chua [3,6]. The hysteresis in measured RRAM I–V curves is a direct result of this.

III. WHY RRAM?

A. The eFlash Replacement Argument

The primary reason RRAM is commercially relevant is that it can replace embedded flash at nodes where flash no longer works. Flash stops scaling around 28 nm for two reasons: the tunnel oxide has a minimum thickness set by retention that does not shrink with the node, and the charge pumps needed to program it grow in area at smaller nodes [5]. RRAM avoids both. The MIM stack is 5–10 nm thick and does not store charge, so retention is governed by the thermal stability of the oxygen-vacancy distribution rather than oxide leakage. Programming voltages are higher than core-logic V_{DD} but an order of magnitude lower than flash, which makes the on-die charge pump correspondingly smaller [1,5].

A second advantage is BEOL integration. The MIM stack goes in the metal layers above the transistors, between two interconnect levels. The front-end logic flow is unchanged, and the memory module adds only one or two extra masks [1]. Embedded flash, by contrast, requires its own front-end module. BEOL placement also opens a path to 3D stacking that planar charge-storage memories do not have.

B. Why VCM Specifically: Comparison with PCM and CBRAM
VCM is the only RRAM family that has reached significant embedded production. Here is why, compared to each alternative.

VCM vs. CBRAM/ECM. CBRAM uses Ag or Cu as the active electrode. Neither belongs in a logic foundry—Cu is a deep-level

recombination center in silicon and a contamination risk even with diffusion barriers, and Ag is not a standard CMOS material at all. Using CBRAM at a logic foundry requires dedicated tooling and protocols that VCM does not need [1,2]. The physics is also a problem: the metallic filament in CBRAM is more sensitive to thermal activation than the oxygen-vacancy filament in VCM, which complicates retention at the 105–150°C operating temperatures required for automotive and IoT [2,3]. CBRAM has found commercial niches at lower temperatures (Adesto at 130 nm, for example) but has not crossed into HKMG-class nodes.

VCM vs. PCM. PCM deserves a careful comparison because it was commercialized more than ten years ago as the foundational technology for Intel–Micron 3D XPoint, and whether VCM is really dominant is a fair question. The short answer is yes, but not because PCM failed. PCM is in active embedded production at STMicroelectronics on a 28 nm FD-SOI automotive platform [5]—it is a real competitor. The reason VCM has the larger embedded market share comes down to current requirements. PCM stores information in the crystalline (low-resistance) versus amorphous (high-resistance) state of a chalcogenide such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$ [9]. To reset a PCM cell—amorphize it—the material has to be heated above its melting temperature ($T_m \approx 600^\circ\text{C}$ for GST) and then quenched rapidly. To set it, the cell is held between the glass transition temperature T_g and T_m long enough for nucleation and growth [9]. That melt-quench requirement drives reset currents to $\sim 100 \mu\text{A}$ or more in scaled cells. A VCM cell resets by partially dissolving an oxygen-vacancy filament without any melting, at currents of $\sim 10\text{--}50 \mu\text{A}$. The higher reset current in PCM forces a wider select transistor, which erodes the cell-area advantage [1,5]. Heat from a PCM cell can also disturb neighboring cells at small pitch, limiting array density. VCM cells dissipate far less energy per switching event and have no such thermal crosstalk [3]. HfO_2 and TaO_x are already qualified in HKMG flows; chalcogenides are not.

PCM retains an advantage when very tight resistance distributions are needed—3D XPoint exploited this for selectorless crosspoint operation. For embedded eFlash replacement at moderate write rates, VCM is the better choice.

VCM vs. STT-MRAM. STT-MRAM is not an RRAM variant, but it is the dominant emerging eNVM at FinFET nodes and the most important external comparison. MRAM has substantially higher endurance ($10^{14}+$ vs. $10^6\text{--}10^9$ cycles for RRAM), no forming step, and deterministic switching. RRAM wins on cell area, mask adder, and the ability to program continuous conductance levels for analog applications. The two technologies coexist: MRAM at sub-22 nm FinFET nodes where endurance and density dominate, RRAM at planar and FD-SOI nodes where mask count, area, and write-rare workloads matter more [5].

C. The Emerging NVM Landscape

By the mid-2020s, embedded RRAM is a foundry option at multiple major fabs. Fig. 5 shows the industry timeline from Hellenbrand *et al.* for FRAM, MRAM, PCRAM, and RRAM from 2017 to 2026 [5]. eMRAM fills the FinFET-class nodes (8–16 nm) at TSMC, Samsung, and GlobalFoundries, while eRRAM

Table 1. Comparison of RRAM and Conventional Memory Technologies

	SRAM	DRAM	NAND	NOR	RRAM
Volatile?	Vol.	Vol.	NV	NV	NV
Cell (F^2)	~ 50	~ 6	~ 4	~ 10	6–12
Read	Fast	Fast	Slow	Fast	Fast
Write	Fast	Fast	Slow	Slow	Med.
Endurance	∞	10^{15}	10^{4-5}	10^{4-5}	10^{4-6} p.
Write V	V_{DD}	V_{DD}	~ 20 V	~ 20 V	~ 2 V
Granularity	bit	bit	block	block	bit
On-chip	Yes	No	No	Yes	Yes

clusters at planar bulk and FD-SOI nodes (22–40 nm). PCM is less represented but in active production.

The headline RRAM deployments are:

- **TSMC.** HfO₂-based embedded RRAM at 40 nm [7], 28 nm, and 22 nm planar bulk, with a 12 nm FinFET demonstration at IEDM 2023 [5]. The 40 nm macro from Chou *et al.* is a 256K×44 array with on-chip current limiting, illustrating directly how cell sizing and peripheral circuit design interact [7].
- **Intel.** Embedded RRAM in the 22FFL FinFET process using TaO_x, with reported endurance of $\sim 10^4$ cycles—a production number, not a lab result [8].
- **Weebit Nano + GlobalFoundries / SkyWater.** TaO_x ReRAM IP in the 22FDX FD-SOI platform and the SkyWater 130 nm bulk platform [5].
- **IBM.** Investigating TaO_x and HfO₂ stacks for analog in-memory compute, using the continuous conductance levels available in VCM cells [5, 6].

Almost all applications are embedded: firmware, calibration constants, AI inference weights, and secure keys in microcontrollers, automotive ICs, IoT chips, and secure elements [5]. Writes are rare, retention requirements are long, and going off-chip for storage is not acceptable for latency or security reasons.

RRAM is not a SRAM replacement. SRAM operates at sub-1 V V_{DD} and at write rates that would exhaust RRAM endurance within hours. RRAM SET requires 1.5–2 V at the bitline and forming requires 3–4 V—both well above the SRAM operating window [5].

D. Position in the Memory Hierarchy

Table I places RRAM directly against the dominant memory technologies. RRAM sits between DRAM and SRAM in cell area. It reads fast, writes at medium speed, and programs per bit. Production endurance is 10^4 – 10^6 cycles for HfO_x and up to 10^{12} in TaO_x lab cells—better than flash but far below DRAM. Write voltage is around 2 V, generated on-chip, much lower than flash. Its niche is the embedded NVM role formerly held by NOR flash, with better scalability and a path to denser arrays.

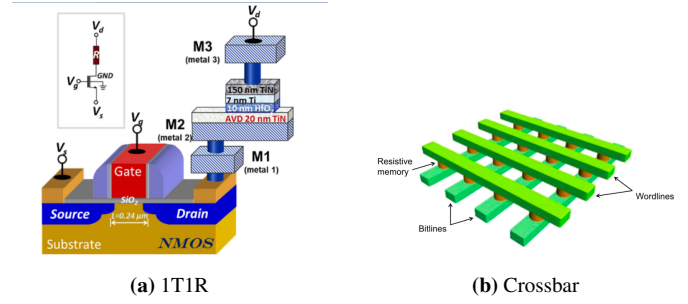


Fig. 5. Competing RRAM array architectures. (a) 1T1R: a CMOS NMOS in the front-end with the HfO₂/Ti/TiN MIM stack between metals 2 and 3. Reproduced from [4]. (b) Passive crossbar. Reproduced from [3].

Table 2. Bias Schemes for 1T1R and Crossbar Arrays

Op.	1T1R	Crossbar
READ	BL: 0.3–0.4 V; SL: 0 V; WL: V_{DD} . MOSFET in triode; sense I_{cell} vs. I_{ref} ; $R_{HRS}/R_{LRS} \sim 10$ – $1000\times$.	WL: $-V_r/2$, BL: $+V_r/2$. Sneak current $(N-1)V_r/(2R_{LRS})$ limits fan-out.
SET	BL: 2–2.5 V; SL: 0 V; WL: V_{DD} . MOSFET in saturation to limit current; $R_{LRS} \approx V_c/I_{cc}$.	WL: $+V_w/2$, BL: $-V_w/2$. Cannot limit current; $N-1$ half-selected cells.
RESET	SL: 1.5 V; BL: 0 V; WL: V_{DD} . MOSFET in saturation (reverse polarity).	BL: $+V_w/2$, WL: $-V_w/2$. $N-1$ half-selected cells.

Compiled from [3, 4].

IV. RRAM IN CMOS: 1T1R INTEGRATION

A. Competing Architectures: 1T1R and the Crossbar

There are two main RRAM array architectures, shown in Fig. 6. In 1T1R, every cell has its own select MOSFET in the front-end, with the RRAM stack in the BEOL above it [4]. In the crossbar, RRAM cells sit at every intersection of orthogonal bitlines and wordlines—no per-cell transistor.

The bias schemes are very different and are summarized in Table II. In 1T1R, the MOSFET clamps cell current and isolates unselected rows. The scheme is straightforward: drive the bitline (BL) to the read or write voltage, hold the source line (SL) at ground, and use the wordline (WL) to gate the MOSFET into triode (read) or saturation (write). In the crossbar, the selected wordline is biased at $+V_w/2$ and the selected bitline at $-V_w/2$, so only the target cell sees the full programming voltage. Every half-selected cell still sees $V_w/2$, however, producing a sneak current:

$$I_{sneak} = \frac{(N-1)V_r}{2R_{LRS}} \quad (1)$$

that limits how large the array can grow before read margin collapses [3]. The crossbar promises a theoretical $4F^2$ cell footprint and a path to 3D stacking, but it only works if sneak current is suppressed by a reliable two-terminal selector at every node. No such selector has reached foundry production for embedded RRAM, which is why every commercial macro today is 1T1R.

Table 3. RRAM Requirements Mapped to Select-FET Design Knobs

RRAM requirement	Transistor design knob
Limit I_{CC} for SET and forming	Operate in saturation; size W/L so that $I_D = I_{CC} = \frac{1}{2}\mu_n C_{ox}(W/L)(V_{GS} - V_T)^2$.
Suppress sneak leakage from unselected cells	Minimize subthreshold swing $SS = (kT/q) \ln 10 (1 + C_d/C_{ox})$; limit DIBL.
Ensure clean read margin	Keep $R_{on} = L/[\mu_n C_{ox} W (V_{GS} - V_T)]$ well below R_{LRS} .

B. Why 1T1R: Mapping RRAM Requirements to Transistor Design

The MOSFET in a 1T1R cell solves three problems that a passive crossbar cannot address. Each maps to a specific transistor design parameter, summarized in Table III.

The most critical requirement is current compliance during SET and forming. Without a current limit, the cell undergoes permanent hard breakdown. The select MOSFET is biased into saturation to act as a current source, and sized so that:

$$I_D = I_{CC} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2 \quad (2)$$

I_{CC} is also a design variable in its own right. Lower compliance produces a thinner filament with higher R_{LRS} , but better endurance and lower switching energy [3]. Fig. 7 shows R_{LRS} and I_{reset} plotted against I_{CC} for several oxide systems—both scale over more than two decades. This is what makes multilevel and analog operation possible. From the MOSFET side, targeting a specific I_{CC} is a W/L sizing problem combined with V_{GS} control through the wordline.

The second constraint is the off-state. In a large array, every unselected cell contributes subthreshold leakage that adds up and degrades read margin. The key figure of merit is subthreshold swing $SS = (kT/q) \ln 10 (1 + C_d/C_{ox})$, which controls how quickly I_D drops below V_T . A steeper swing means more cells can share a wordline without the cumulative leakage corrupting a read. This is one reason FinFETs and FD-SOI MOSFETs—which have smaller C_d/C_{ox} —are attractive hosts for RRAM. DIBL matters too, because it shifts V_T at high V_{DS} , which changes the saturation I_D used to set I_{CC} .

The third constraint is read margin. The on-resistance $R_{on} = L/[\mu_n C_{ox} W (V_{GS} - V_T)]$ of the select transistor must stay well below R_{LRS} —otherwise the read voltage divides between the cell and the transistor and the HRS/LRS distinction collapses.

C. A Worked Example: Niu 2016 Planar Process

Niu *et al.* give a concrete sizing example on a 0.25 μm planar CMOS line [4]. The cell is a $\text{HfO}_2/\text{Ti}/\text{TiN}$ MIM stack between metals 2 and 3 above an NMOS. The series circuit gives:

$$V_{BL} = V_{RRAM} + V_{DS} \quad (3)$$

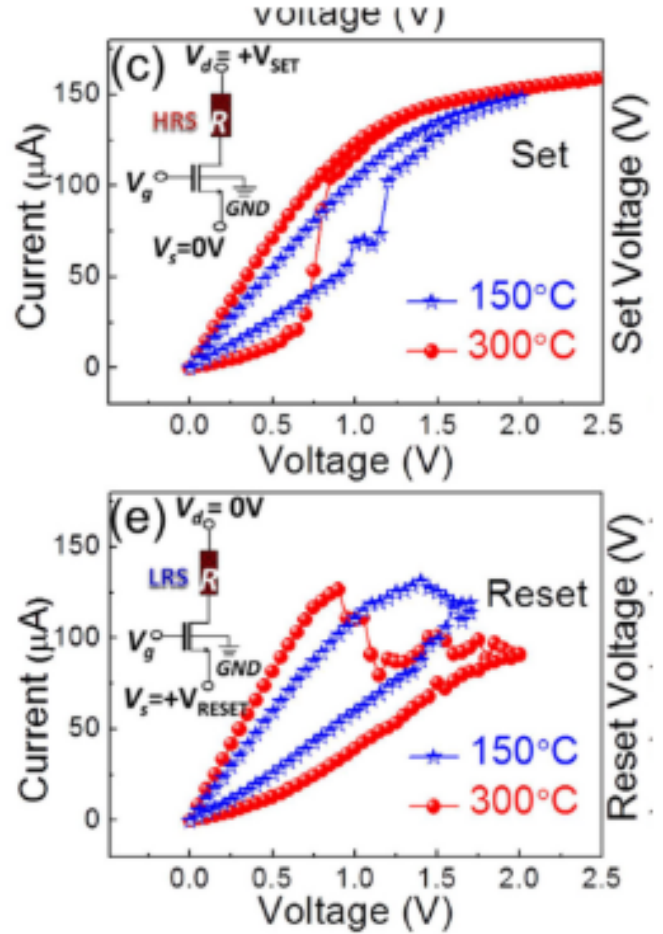


Fig. 6. Measured I-V of the 1T1R HfO_2 cell from [4] at two ALD temperatures (150°C blue, 300°C red). Top: SET; Bottom: RESET. Higher temperature produces tighter distributions and lower switching voltages. Reproduced from [4].

For the MOSFET to stay in saturation during SET, the condition is $V_{BL} - V_{SET} > V_{GS} - V_T$. With $W = 1.14 \mu\text{m}$ and $L = 0.24 \mu\text{m}$ at $V_g = 0.9 \text{ V}$, Eq. (2) gives $I_{CC} = 10 \mu\text{A}$. From that, $R_{LRS} = V_c/I_{CC} = (0.4 \text{ V})/(10 \mu\text{A}) = 40 \text{ k}\Omega$, using $V_c \approx 0.4 \text{ V}$ for HfO_2 . Transistor on-resistance from the saturation model is $R_{on} \approx 200 \Omega$ —well below the 40 k Ω LRS, satisfying the read-margin constraint. Forming needs 3.2–4.0 V at the bitline. After forming, SET drops to 0.8–1.1 V and RESET to 1.0–1.5 V [4].

The I-V curves for this cell at two ALD deposition temperatures are shown in Fig. 8. Higher deposition temperature reduces residual carbon impurities from the metal-organic precursor and tightens the resistance distributions, lowering both V_{SET} and V_{RESET} [4]. The broader point is that RRAM electrical performance is inseparable from process choices in the materials stack.

V. RELIABILITY AND DISTURB MECHANISMS

A. Endurance and the Fresh-Oxide Question

A reasonable question for any NVM is whether each RESET fully restores the cell to its as-fabricated state. For RRAM, the answer is no—and the distinction matters. Forming happens once and nucleates the filament path. Every SET and RESET after that reorganizes oxygen vacancies along that same path. Over cycling, oxygen ions accumulate at electrode interfaces, lattice strain builds up, and some defects become permanently fixed in positions no SET pulse can move [3]. Endurance is the number of cycles before the LRS–HRS read window narrows enough that a comparator can no longer tell the two states apart.

Production HfO_x cells reach 10^4 – 10^6 cycles. Laboratory TaO_x cells have demonstrated up to 10^{12} cycles in single-device tests [3, 5]. The gap between those numbers suggests the ceiling is materials- and process-specific rather than fundamental. Two failure modes dominate: R_{LRS} drifts up when the filament cannot fully reconnect after RESET, or R_{HRS} drifts down when residual conductive paths persist. Either way the read window closes. Endurance and retention are coupled to the same process variables—oxide stoichiometry, electrode work function, getter layer thickness—which is why they are co-optimized together [3, 4].

B. Retention and Disturb Mechanisms

Retention is a claim about how long the programmed resistance distribution stays stable at operating temperature. LRS retention is governed by Arrhenius-activated dissolution of the filament, with activation energies of 1.0–1.5 eV for HfO_x [3]. The HRS distribution is not static either: even small read voltages slowly redistribute oxygen vacancies and broaden the distribution. This is called read disturb.

Fig. 9 shows both effects in a HfO_2 1T1R cell. Both distributions broaden with time, faster at higher temperature [3]. The HRS tail drifts toward lower resistance, narrowing the read window from above. A cell that reads clearly in HRS at 25°C may have drifted into the uncertain region at 125°C after a few years—which is why elevated-temperature bake tests are a standard part of RRAM qualification.

In 1T1R, write disturb on neighboring cells is small because the select FET isolates each row, and read disturb is controlled because the read voltage (0.3–0.4 V) stays well below the SET threshold. The main disturb mechanism in 1T1R is the gradual drift shown in Fig. 9—slow thermal degradation over time and temperature.

The crossbar has an additional, more acute disturb problem. Every write operation stresses every cell on the active wordline and bitline at $V_w/2$ —not just the target cell. Over many writes, this repeated half-voltage stress accumulates and drives the same Arrhenius degradation that in 1T1R builds only gradually. Fig. 9 illustrates what is at stake: the same resistance drift that is slow and manageable in 1T1R becomes a repeated, forced excitation of half-selected cells in the crossbar. A strongly non-linear selector is therefore not optional in a crossbar—without it, half-selected cells degrade during every normal write operation.

Table 4. Comparison of MOSFET Architectures Hosting RRAM

Parameter	Planar bulk	FD-SOI	FinFET
V_{DD}	1.0 V	0.9–1.0 V	0.8–0.9 V
I_{CC} control	$W/L + V_g$	$W/L + V_g + \text{BG}$	$N_{\text{fin}} + V_g$
SS (mV/dec)	80–100	65–75	~65
Min. cell area	$0.06 \mu\text{m}^2$	$0.066 \mu\text{m}^2$	$0.022 \mu\text{m}^2$
Forming headroom	OK	pump may be needed	pump required
Production eRRAM	TSMC 28 nm	STMicro/GF 22FDX	TSMC 12 nm

Adapted from [3, 5].

C. Voltage Envelope and Charge-Pump Implications

A practical question for any embedded NVM is whether the programming voltages can be generated on-chip. For RRAM: forming needs 3.2–4.0 V at the bitline; SET and RESET need 1.5–2 V; read needs 0.3–0.4 V [4]. At 10–50 μA per cell, even a kilobit-scale programming operation draws on the order of milliamps. Both the voltage level and the current are within what an on-die switched-capacitor charge pump can deliver. Every shipping embedded RRAM macro generates its own high voltages internally—no off-chip supply is needed. Flash needs 15–20 V and requires a much larger on-die pump network [1, 5]. This is part of why RRAM’s mask adder ends up smaller than flash even though the cell stack is comparable in complexity.

VI. TECHNOLOGY-DEPENDENT CHALLENGES: PLANAR, FD-SOI, AND FINFET

The RRAM cell is the same across architectures, but the select-FET problem changes. Cell density, I_{CC} control precision, and forming headroom all depend on which MOSFET platform is used. Table IV summarizes the comparison.

Planar bulk (250 nm down to 28 nm). This is the baseline. Continuous W and L sizing gives the most flexibility for setting I_{CC} . The penalty is the worst subthreshold slope of the three options—80–100 mV/dec—which limits array fan-out. TSMC’s 40 nm and 28 nm products are on planar bulk [5, 7], as is the Niu 2016 case study.

FD-SOI (Samsung 28FDSOI, GlobalFoundries 22FDX). The fully-depleted SOI transistor adds an independent back-gate. In the 1T1R cell, this provides per-die threshold-voltage trim of the select FET, translating directly into per-die I_{CC} trim—useful for compensating both process variation in the transistor and the larger device-to-device variation in the RRAM cell. Subthreshold slope improves to 65–75 mV/dec because the buried oxide removes the body capacitance term C_d . The tradeoffs are a slightly larger minimum cell area and tighter forming headroom, since V_{DD} stays around 1 V while the forming voltage is still near 3 V. Weebit Nano’s TaO_x IP runs on the 22FDX platform [5].

FinFET (16 nm and below). FinFETs give the steepest subthreshold slope (~65 mV/dec), best leakage suppression, and highest drive current per footprint. Cell area drops to $\sim 0.022 \mu\text{m}^2$. The challenge is that W is quantized into integer fin counts. Targeting a specific I_{CC} may require rounding to the nearest fin, producing a discrete I_{CC} ladder rather than a smooth curve. The lower V_{DD} (0.8–0.9 V) also leaves less headroom for the forming voltage, making a dedicated charge

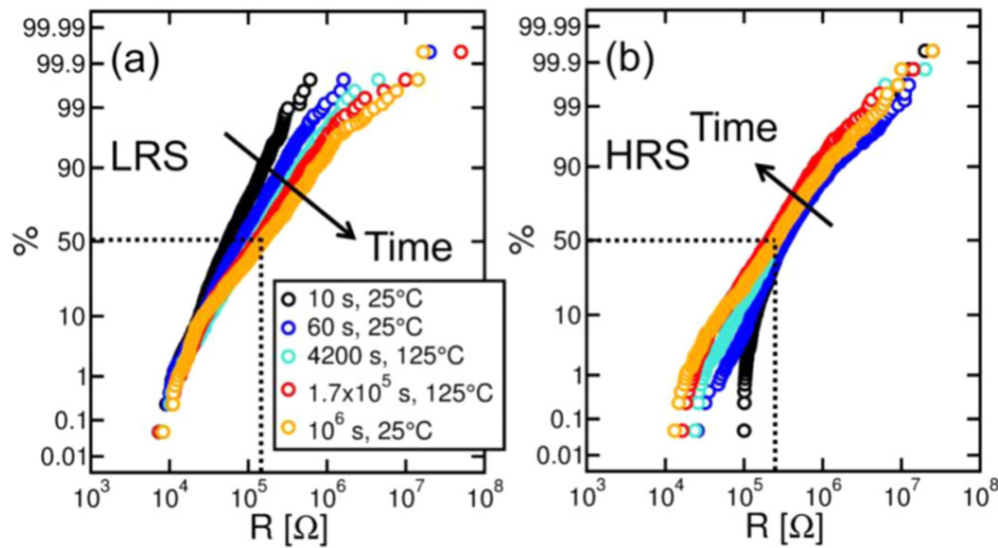


Figure 20. Statistical study of retention for LRS and HRS. Both LRS (a) and HRS distributions (b) indicate broadening with time. Even if the value of HRS resistance is forced above $10^5 \Omega$, the distribution soon extends to lower resistance values due to noise. Copyright (2015) IEEE. Reprinted, with permission, from [109].

Fig. 7. Statistical retention study for LRS (a) and HRS (b) distributions in a HfO₂ 1T1R cell at multiple temperatures and times. Both distributions broaden with time; the HRS tail drifts toward lower resistance and is the dominant read-disturb signature. Reproduced from [3].

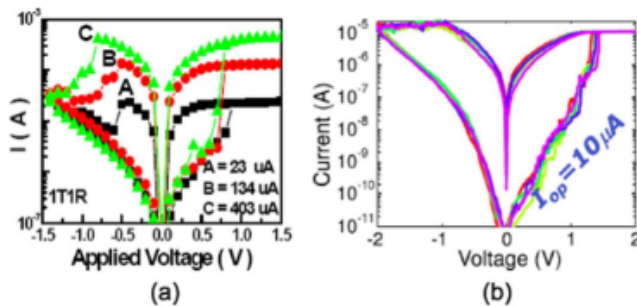


Fig. 8. Measured I–V of an integrated 1T1R HfO_x cell at three compliance current levels (a), and resulting LRS/HRS distributions vs. compliance (b). The compliance current is set by the select MOSFET. Reproduced from [3].

pump effectively mandatory. TSMC demonstrated embedded RRAM on 12 nm FinFET at IEDM 2023 [5].

Fig. 10 shows how cell characteristics depend on I_{CC} across all three architectures. The select MOSFET sets compliance, and R_{LRS} , I_{reset} , and read margin all follow it. The architecture choice affects not just area and fan-out but the precision of the I_{CC} knob—continuous on planar, continuously trimmable on FD-SOI, and quantized on FinFET.

VII. CONCLUSION AND OUTLOOK

RRAM is a serious contender to replace embedded flash alongside STT-MRAM and PCM. Within RRAM, the industry settled on VCM using HfO₂ or TaO_x. The reasons are concrete: both oxides are already in the fab, the oxygen-vacancy filament is thermally more stable than what CBRAM produces, and the reset current is an order of magnitude lower than PCM's melt-quench requirement. PCM is not a failed technology—it is in production today—but VCM has taken the larger embedded market share for those specific, practical reasons.

The MOSFET content of this technology is concentrated in the select transistor. Three requirements—current compliance during SET and forming, leakage suppression from unselected cells, and read margin—map directly onto three transistor parameters: W/L sizing in the saturation model, subthreshold swing and DIBL, and R_{on} . The Niu 2016 sizing example on a 0.25 μm process makes this mapping concrete. Moving to FD-SOI adds a back-gate for per-die I_{CC} trim. Moving to FinFET cuts cell area and improves leakage but quantizes the sizing knob and tightens forming headroom. Each step on the MOSFET roadmap reshapes the select-FET problem rather than eliminating it.

The most interesting forward-looking application for RRAM is not as a memory replacement but as the storage element in an analog matrix-vector multiplier for AI inference. A VCM cell's conductance can be set continuously by controlling I_{CC} during SET. An array of such cells physically implements the dot

products at the core of neural-network inference directly inside the memory, bypassing the von Neumann bottleneck [6]. Yao *et al.* demonstrated a fully hardware-implemented memristor convolutional neural network on a CMOS-compatible RRAM array, reaching accuracy comparable to a software baseline on standard image benchmarks [10]. If RRAM finds a role beyond embedded NVM, this is the most plausible one—not as DRAM or SRAM replacement, but as a fabric where memory and computation happen in the same physical device.

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